CHAPTER-1

INTRODUCTION

1.1 WOUND CARE AND HEALING

Wound care is a critical aspect of healthcare that focuses on preventing infection, promoting tissue regeneration and ensuring faster recovery of damaged skin. Proper wound management becomes especially important in cases of cut wounds and burns. The natural healing process of the human body involves a series of stages including hemostasis, inflammation, proliferation and tissue remodeling. However improper care, microbial contamination and delayed treatment can significantly slow down this process, leading to complications such as infection, prolonged pain and tissue damage. Traditional wound dressing methods primarily involve the use of cotton gauze, antiseptic solutions and bandages. While these methods provide basic protection, they often lack the ability to actively support healing or monitor the condition of the wound. Frequent manual inspection is required to assess healing progress, which may lead to delayed detection of infections or improper wound conditions. In many cases, conventional dressings fail to maintain optimal moisture levels, which is essential for effective tissue regeneration. In recent years, the integration of natural medicinal compounds in wound treatment has gained attention due to their antimicrobial, anti inflammatory, and healing properties. Medicinal plants such as Tridax procumbens have shown significant potential in accelerating wound healing by promoting cell proliferation and reducing microbial growth. However, their application in a structured, controlled and user friendly form remains limited. Therefore, there is a growing need for an advanced wound care solution that not only protects the wound but also enhances healing through medicinal support, maintains a suitable microenvironment and reduces the dependency on frequent manual intervention.

The development of an intelligent and efficient wound dressing system can significantly improve patient outcomes, especially in resource limited settings. In addition to these challenges, the increasing prevalence of chronic wounds such as diabetic ulcers, pressure sores and post-surgical wounds has further highlighted the need for improved wound care solutions. Patients with conditions like diabetes or compromised immunity often experience delayed healing, making them more vulnerable to infections and complications. This not only affects their quality of life but also increases. the burden on healthcare systems due to prolonged treatment and hospitalization. Moreover, the demand for eco friendly and biocompatible materials in biomedical applications is steadily rising.

Synthetic wound dressings, though effective in some cases, may cause skin irritation, allergic reactions, and environmental concerns due to non-biodegradability.

In contrast, natural polymers and plant based extracts offer a safer and more sustainable alternative, as they are biodegradable, less toxic, and more compatible with human tissues.Another important aspect of modern wound care is the need for materials that can maintain an optimal moist environment, allow proper gas exchange and provide a barrier against external contaminants. Advanced wound dressings are now being designed to actively participate in the healing process rather than just acting as passive coverings. Incorporating bioactive compounds into such dressings can significantly enhance their functionality by promoting faster cell regeneration and reducing inflammation.Furthermore, accessibility and affordability remain key concerns, especially in rural and low resource settings where advanced medical facilities may not be readily available. Developing a low cost, easy to use, and effective wound dressing can bridge this gap and provide better healthcare solutions to a wider population. Therefore, integrating natural medicinal agents with simple fabrication techniques presents a promising approach toward achieving efficient, safe and widely accessible wound care systems.

1.2 EXISTING SOLUTION

At present wound care systems mainly rely on conventional dressing techniques using sterile gauze, adhesive bandages and antiseptic creams. In hospitals, advanced wound care may include hydrogel dressings, antimicrobial dressings, and periodic clinical monitoring. These methods are effective to a certain extent but still depend heavily on manual observation and timely replacement of dressings.Some modern wound care solutions incorporate medicated dressings infused with antibiotics or synthetic healing agents. However, these solutions are often expensive and may not be easily accessible to all patients. Additionally, the overuse of chemical based treatments can sometimes lead to skin irritation or antibiotic resistance.Most current systems do not provide real time monitoring of wound conditions such as moisture, infection level, or healing progress. Patients and caregivers must rely on visual inspection, which can be subjective and inaccurate. This limitation makes early detection of complications difficult, particularly in home care settings. In addition to these limitations, many existing wound care systems lack personalization and adaptability to different types of wounds. Each wound varies in size, depth, and severity, yet conventional dressings are often designed as “one size fits all” solutions. This can result in inadequate protection or inefficient healing for certain conditions.

Furthermore, frequent removal and replacement of dressings can disturb newly formed tissues, causing delayed healing and in severe cases, further tissue damage or systemic complications. Maintaining a sterile and controlled environment is challenging with conventional approaches.Additionally, the high cost of advanced treatments presents a barrier for many patients. Modern wound care solutions such as hydrogel dressings, antimicrobial dressings, and specialized clinical treatments offer improved healing but are often expensive. These costs can be difficult to afford, particularly for patients in low income groups or in developing regions, limiting their widespread use.The issue of limited accessibility further compounds the problem. Advanced wound care facilities and products are usually concentrated in urban hospitals and specialized healthcare centers. Patients living in rural or remote areas may not have access to such facilities, leading to reliance on basic and less effective treatment methods.

The gap in healthcare access can significantly impact patient outcomes.Finally, there is the challenge of inconsistent healing conditions. Effective wound healing requires an optimal environment, including proper moisture balance, temperature and protection from external factors. Traditional dressings often fail to maintain these conditions consistently. For example wounds may become too dry or excessively moist, both of which can hinder the healing process. Without a stable micro environment, tissue regeneration becomes slower and less efficient.Overall, these limitations highlight the need for improved wound care solutions that not only protect the wound but also actively support healing, reduce dependency on manual monitoring and are affordable and accessible to a wider population.

1.3 PROPOSED SOLUTION

The proposed system presents an advanced medicated wound dressing that integrates traditional herbal medicine with modern biomedical engineering to improve the efficiency of wound healing. This system is specifically designed to support and enhance the body's natural repair mechanisms by incorporating bioactive plant-based compounds, particularly derived from Tridax procumbens, a well-known medicinal herb recognized for its wound healing properties.

Wound healing is a complex physiological process involving multiple overlapping stages, including hemostasis, inflammation, proliferation, and remodeling. Any disruption in these phases can delay recovery and increase the risk of infection. Conventional wound dressings primarily function as passive

barriers that protect the wound from external contaminants but do not actively participate in the healing process. In contrast, the proposed medicated dressing acts as an active therapeutic system that accelerates healing by delivering natural bioactive compounds directly to the wound site.

The herbal extract of Tridax procumbens plays a vital role due to its proven antimicrobial, anti-inflammatory, and antioxidant properties. The antimicrobial activity helps inhibit the growth of harmful microorganisms such as bacteria and fungi, thereby reducing the chances of infection. Its anti-inflammatory effects help in minimizing swelling, redness, and pain, creating a more favorable environment for tissue repair. Additionally, the presence of antioxidants supports cellular regeneration by neutralizing free radicals, which can otherwise damage healthy cells and delay healing.

Another key feature of this system is its ability to maintain an optimal moist environment at the wound site. Moist wound healing is widely recognized as an effective approach because it promotes faster cell migration, enhances collagen synthesis, and reduces scar formation. The dressing is engineered to balance moisture retention while preventing excessive exudate accumulation, thereby ensuring ideal conditions for tissue regeneration.

The structural design of the dressing focuses on ease of application and user convenience. It can be directly applied to various types of wounds such as cuts, abrasions, and minor burns, making it suitable for both clinical and home care use. Its flexibility and adherence ensure proper coverage of the wound without causing discomfort to the patient.

Furthermore, the use of biocompatible and biodegradable materials ensures that the dressing is safe for human skin and does not cause irritation or allergic reactions. Unlike synthetic chemical-based wound care products, this herbal-based system significantly reduces the risk of side effects and antibiotic resistance, which is a growing concern in modern healthcare. The environmentally friendly nature of biodegradable materials also contributes to sustainable healthcare practices.

The simplicity of preparation and cost-effectiveness make this system particularly beneficial for rural and low-resource settings where access to advanced medical treatments is limited. By utilizing locally available medicinal plants and simple fabrication techniques, this dressing can be produced at a low cost without compromising its effectiveness.

The proposed wound dressing system offers an effective and natural approach to wound care by utilizing the healing properties of Tridax procumbens. It enhances faster recovery while reducing the risk of infection and complications. The system is safe, cost-effective, and suitable for use in both hospitals and home settings.

1.4 SCOPE OF THE SYSTEM

The Nature Boosted Wound Healing using Tridax procumbens with collagen is designed to provide an advanced and natural solution for wound care. The scope of this system focuses on developing a bioactive wound dressing that not only protects the wound physically but also enhances the healing process biologically. Conventional bandages mainly act as protective coverings and often lack antimicrobial and healing promoting properties. This project addresses that limitation by incorporating the medicinal plant Tridax procumbens, which is well known for its antibacterial, anti-inflammatory, and wound healing capabilities.

The inclusion of collagen in the system expands its scope further by providing a biocompatible and biodegradable matrix that supports cell growth and tissue regeneration. The band is designed to maintain a moist wound environment, which is essential for faster healing and reduced scar formation. Additionally, the system aims to reduce the use of synthetic antiseptics by offering a natural alternative that minimizes side effects and improves patient safety.

The application scope of the Tricol BioRegen Band includes treatment of minor cuts, burns, abrasions, and basic wound care. With further development, it can be extended to more complex applications such as diabetic wound management and clinical wound care systems. The project also opens future possibilities for integrating drug delivery systems and smart wound monitoring technologies, making it a promising innovation in the biomedical field.

1.5 ORGANIZATION OF THE SYSTEM

The organization of the Tricol BioRegen Band - Nature Boosted Wound Healing using Tridax procumbens with collagen follows a systematic and structured approach to ensure effective development of the wound dressing. Initially, the process begins with raw material preparation, where *Tridax procumbens* leaves are collected, cleaned, dried, and processed to extract their bioactive compounds. At the same time, collagen is prepared as a biocompatible base material. These components are then combined in controlled proportions to form a uniform bioactive composite, ensuring proper consistency and therapeutic effectiveness. The mixture is subsequently cast into thin films and dried under controlled environmental conditions to obtain the final band structure.

After fabrication, the system proceeds to the evaluation and testing phase to ensure quality, safety, and performance.Various physical, chemical, and biological tests are carried out, including thickness measurement, tensile strength analysis, pH evaluation, cytotoxicity testing, and hemocompatibility studies. Antimicrobial activity is also assessed to confirm the band's ability to prevent infection. Finally, the prepared band is applied and observed for its wound healing efficiency, and the results are analyzed and compared with conventional bandages. This organized workflow ensures that the developed system is reliable, effective and suitable for practical biomedical applications

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

Wound healing is a complex and dynamic biological process essential for restoring the structural and functional integrity of damaged tissues. It involves a highly coordinated sequence of cellular, molecular and biochemical events that occur in a regulated manner to ensure effective tissue repair. This process is generally classified into four overlapping phases: hemostasis, inflammation, proliferation and remodeling. Each phase is controlled by specific signaling pathways, cellular responses and biochemical interactions that collectively determine the efficiency and outcome of wound healing. Any imbalance or disruption in these phases can result in delayed healing, chronic wounds or excessive scar formation, highlighting the need for effective biological interventions.

The initial phase of wound healing, known as hemostasis, occurs immediately after tissue injury and is characterized by vasoconstriction and platelet aggregation. Platelets release clotting factors and form a fibrin matrix that serves as a temporary extracellular scaffold. This matrix not only prevents excessive blood loss but also provides a structural framework for cell migration during subsequent healing phases. Following hemostasis, the inflammatory phase is initiated, during which immune cells such as neutrophils and macrophages infiltrate the wound site. These cells play a critical role in eliminating pathogens, clearing cellular debris and releasing proinflammatory cytokines and growth factors that regulate the healing process. Although inflammation is essential for initiating tissue repair, prolonged or excessive inflammatory responses can lead to tissue damage and delayed healing.The proliferative phase represents a critical stage in wound healing, involving fibroblast proliferation, collagen synthesis, angiogenesis and reepithelialization. Fibroblasts are the primary cells responsible for the synthesis of extracellular matrix components, particularly collagen, which provides structural strength to the newly formed tissue. Angiogenesis, the formation of new blood vessels, ensures an adequate supply of oxygen and nutrients required for cellular metabolism and tissue growth.

Wound healing represents a highly intricate and dynamic biological phenomenon that is fundamental to the restoration of tissue integrity following injury. It is not merely a passive response, but a precisely regulated cascade of cellular, molecular, and biochemical events that interact in a coordinated manner to ensure effective repair and regeneration. Over the years, extensive research has demonstrated that successful wound healing depends on the seamless progression of multiple overlapping phases, namely hemostasis, inflammation, proliferation, and remodeling. Each of these phases is governed by specific signaling mechanisms and cellular activities, which collectively influence the overall quality and speed of tissue recovery. Any disruption in this delicate balance may result in impaired healing outcomes, including chronic wounds or abnormal scar formation, thereby emphasizing the importance of controlled biological interventions.

The process is initiated immediately after injury with the hemostatic phase, a critical defensive mechanism aimed at preventing excessive blood loss. This phase is characterized by rapid vasoconstriction and platelet aggregation, leading to the formation of a fibrin clot. Beyond its primary role in coagulation, this fibrin matrix serves as a provisional extracellular scaffold that supports the migration of various cells required for subsequent healing stages. As hemostasis stabilizes the wound environment, the inflammatory phase follows, marked by the infiltration of immune cells such as neutrophils and macrophages. These cells are essential for host defense, as they actively remove microbial contaminants and cellular debris while simultaneously releasing cytokines and growth factors that orchestrate further healing responses. However, it is important to recognize that while inflammation is indispensable, its persistence beyond the required duration can be detrimental, contributing to delayed healing and tissue damage.

Progressing from inflammation, the proliferative phase plays a pivotal role in rebuilding the damaged tissue architecture. This stage is characterized by active fibroblast proliferation, synthesis of extracellular matrix components, and the deposition of collagen, which provides mechanical strength to the newly formed tissue. Concurrently, angiogenesis ensures the development of new blood vessels, thereby supplying essential nutrients and oxygen to support cellular activities.

2.2 LITERATURE SURVEY ON TRIDAX PROCUMBENS AND GELATIN

Jeevitha et al. (2024) and **Devi et al. (2025)** investigated the wound healing and tissue regeneration potential of *Tridax procumbens*, a medicinal herb rich in phytoconstituents such as flavonoids, tannins, alkaloids, carotenoids and terpenoids. Their studies involved phytochemical analysis and experimental wound healing models. The results demonstrated enhanced fibroblast proliferation, collagen deposition, epithelialization, and accelerated wound contraction, along with strong antimicrobial, antinflammatory, antioxidant and hemostatic properties. However, these studies highlight a gap in terms of limited clinical trials and lack of standardized formulations for human therapeutic use.[1]

Devi et al. (2025) examined the antioxidant and angiogenic properties of *Tridax procumbens* in wound healing. The methodology included evaluation of flavonoids and phenolic compounds for reactive oxygen species (ROS) scavenging, along with analysis of vascular endothelial growth factor (VEGF) and lysyl oxidase activity. The results showed effective reduction of oxidative stress, protection of tissues from damage, and promotion of angiogenesis through VEGF upregulation and collagen stabilization. However, further long term studies and detailed investigation of molecular pathways in human systems are required.[2]

Chinnappan et al. (2023) and **Cheriyan et al. (2025)** investigated the synergistic effects of combining *Tridax procumbens* with polymeric and collagen based matrices. Their methodology included the development of extract loaded films and nanofibrous scaffolds, followed by antimicrobial testing against pathogens such as *Staphylococcus aureus* and *Escherichia coli*. The results demonstrated enhanced antibacterial activity, improved cell viability, increased fibroblast proliferation and superior tissue regeneration. However, gaps remain in terms of scalability, clinical validation and optimization of biomaterial compositions for commercial use.[3]

Zhang et al. (2023) investigated the application of collagen based hydrogels in skin wound healing. The methodology included preparation of collagen hydrogels and evaluation of their structural properties, crosslinking behavior, and biological performance. The results demonstrated that collagen hydrogels provide a three dimensional network structure that enhances tissue repair, supports cell growth and accelerates wound healing. Despite these advantages, challenges remain in optimizing hydrogel properties such as mechanical strength, degradation rate and stability for clinicalapplications.[4]

Cao et al. (2025) explored the role of lyophilization in the fabrication of collagen based scaffolds for tissue engineering. The methodology involved freezing collagen solutions followed by freeze-drying to remove water and create porous structures. The results demonstrated that lyophilization produces highly porous scaffolds (up to 96% porosity), which enhance cell infiltration, nutrient transport, and tissue regeneration. Additionally, pore size and mechanical properties can be controlled by adjusting processing conditions. However, limitations include high processing cost, time consumption and the need for optimization of structural uniformity for clinical applications.[5]

Wu et al. (2025) developed a composite hydrogel combining gelatin, recombinant type III collagen and carboxymethyl cellulose for enhanced wound healing. The method involved chemical crosslinking and in vitro and in vivo evaluation of wound repair efficiency. The results showed improved cell migration, enhanced macrophage response, accelerated wound closure, and better collagen organization in tissue regeneration. However, further research is needed to validate long term clinical performance and scalability of such composite biomaterials.[6]

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

The methodological framework of this study is systematically designed to develop and evaluate a bioactive wound dressing system, namely the Tricol Bioregen Band, by integrating principles of biomaterial science and phytochemical engineering. The primary objective of this methodology is to create a wound healing system that not only provides physical protection but also actively promotes tissue regeneration, prevents microbial infection, and maintains an optimal physiological environment for healing. This approach is based on a structured sequence of processes, including raw material selection, phytochemical extraction, biomaterial preparation, composite formulation and functional evaluation of the developed dressing.

Wound healing is a highly complex biological process involving multiple interdependent phases such as hemostasis, inflammation, proliferation and remodeling. Each phase is governed by precise cellular and biochemical interactions, including platelet aggregation, inflammatory mediator release, fibroblast proliferation, collagen synthesis, and extracellular matrix (ECM) formation. For effective healing, the wound environment must be carefully maintained with adequate moisture, oxygen exchange, and protection from external contaminants. Conventional wound dressings are often inadequate in supporting these dynamic requirements, as they primarily function as passive barriers without actively participating in the healing process. This limitation necessitates the development of advanced wound care systems that can interact at the cellular and molecular levels to enhance healing outcomes.

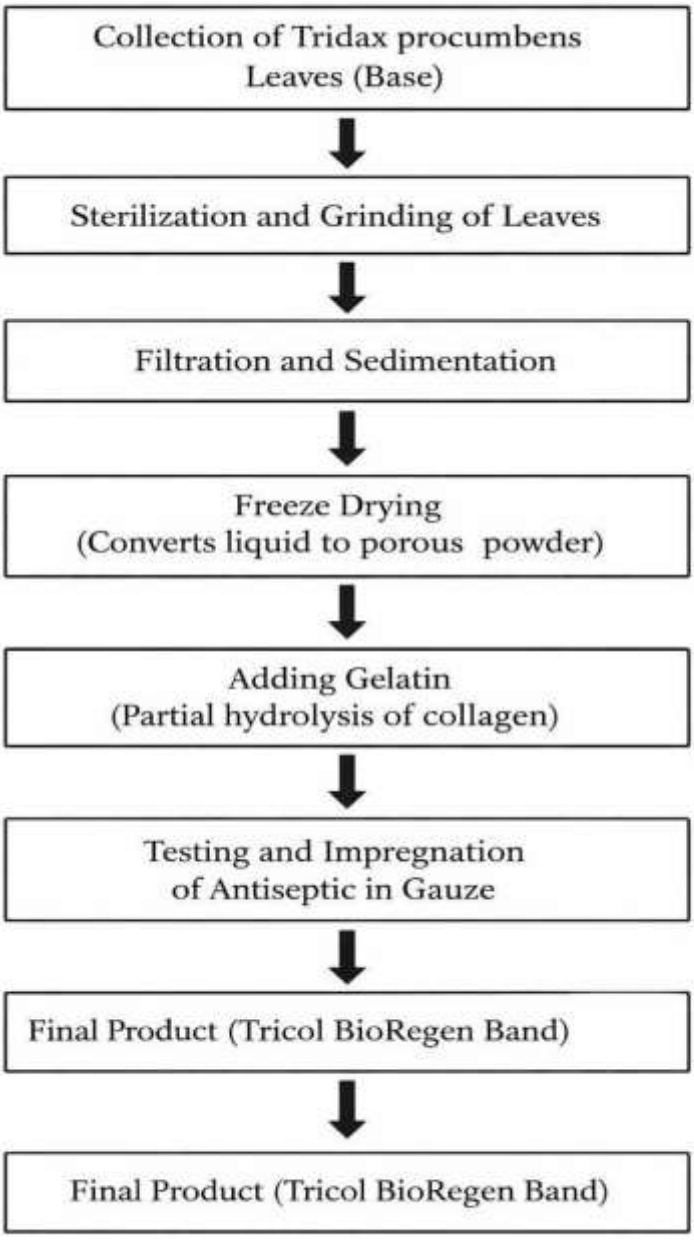
In this context, Tridax procumbens is selected as the primary bioactive component due to its well established medicinal properties. The plant contains a wide range of phytochemicals, including flavonoids, alkaloids, tannins, and carotenoids, which exhibit antimicrobial, antiinflammatory, antioxidant and hemostatic activities. These compounds play a significant role in reducing microbial load, controlling inflammation and promoting fibroblast activity, thereby accelerating tissue regeneration and wound closure. However, the effectiveness of these bioactive compounds

on the efficiency of the extraction and purification process, which must ensure maximum retention of therapeutic constituents without degradation. The extraction process adopted in this study involves a series of carefully controlled steps to obtain a high purity phytochemical extract. Initially, fresh plant material is cleaned and dried under controlled temperature conditions to prevent degradation of heat sensitive compounds. The dried material is then pulverized into fine powder and subjected to solvent based extraction to isolate the active constituents. The extract is further purified through filtration to remove insoluble impurities, followed by sedimentation to separate residual particulate matter. Finally ,freeze drying (lyophilization) is employed to remove moisture under low temperature and vacuum conditions, resulting in a stable and concentrated extract. This process ensures the preservation of the chemical structure and biological activity of the phytoconstituents.

Simultaneously, the biomaterial component of the system is developed using gelatin, which is derived from collagen. Gelatin is widely recognized for its excellent biocompatibility, biodegradability and ability to mimic the natural extracellular matrix. When dissolved in an aqueous medium at controlled temperature, gelatin forms a homogeneous viscous solution capable of encapsulating bioactive compounds. This property allows for the effective incorporation of the Tridax procumbens extract into the gelatin matrix, resulting in a composite material that combines structural support with biological functionality. The interaction between gelatin and the phytochemical extract enhances the stability and controlled release of therapeutic agents at the wound site.

In conclusion, the methodology adopted in this study represents a comprehensive and scientifically rigorous approach to the development of a bioactive wound dressing system. By integrating natural phytochemical therapy with biomaterial engineering, the proposed system offers a cost effective, eco friendly and clinically relevant solution for wound care. The structured methodology ensures reproducibility, scalability and practical applicability, making it suitable for implementation in both advanced healthcare facilities and resource limited settings.

3.2 BLOCK DIAGRAM



3.3 METHODOLOGY OVERVIEW

The preparation of the film involves a series of simple yet important steps to ensure a uniform and effective final product. Initially, gelatin and PVA solutions are prepared separately and then mixed thoroughly to form a stable polymer base. This mixture is allowed to cool and stabilize before the addition of Tridax procumbens extract, which provides the bioactive properties. The prepared solution is then cast into films using proper lubrication to ensure smooth spreading. Following this, the film is dried carefully to achieve proper formation and consistency. Finally, the dried film is applied onto a bandage for practical use.

Table 1:

Measurement of Materials

Material	Quantity
Gelatin	1.6 g (8%)
Polyvinyl Alcohol (PVA) (Plasticizer)	0.6 g (3%)
Tridax procumbens powder	0.2 g
Distilled Water (for Gelatin)	6 ml
Distilled Water (for PVA)	8 ml

3.4 PREPARATION OF POLYMER SOLUTIONS

3.4.1 Gelatin Solution:

Gelatin (1.6 g) was added to 6 ml of distilled water and heated gently at 35°C–40°C. Continuous stirring helped the gelatin dissolve completely, forming a smooth and slightly viscous solution.This step is important because incomplete dissolution can affect the quality of the final film.

3.4.2 PVA Solution:

PVA (0.6 g) was dissolved in 8 ml of distilled water under the same temperature conditions. Since PVA takes slightly longer to dissolve, continuous stirring was maintained until a clear Solution.

3.4.3 Formation of Polymer Matrix

The gelatin and PVA solutions were then mixed together slowly while stirring continuously. This allows the two polymers to interact and form a stable base.At this stage, the solution becomes more viscous, indicating proper blending. The mixture was maintained at a moderate temperature to ensure smooth interaction between the polymers.

3.4.4 Cooling and Stabilization

After mixing, the solution was allowed to cool naturally to room temperature. This step helps stabilize the mixture and prepares it for the addition of the plant extract.Cooling also prevents damage to the bioactive compounds added in the next step.

3.4.5 Addition of Tridax procumbens

Once the solution had cooled, 0.2 g of Tridax procumbens powder was added. The mixture was stirred thoroughly to ensure even distribution.This step is crucial because uniform distribution ensures that the entire film has consistent antimicrobial and healing properties.

3.4.6 Drying and Film Formation

The solution was left undisturbed at room temperature for drying. As the water evaporates, a thin solid film is formed.This step requires patience, as uneven drying can lead to cracks or irregularities. A stable environment ensures better film quality.

3.4.7 Application on Bandage

After drying, the film was gently removed using forceps and placed onto a sterile bandage. The film adhered well due to the natural properties of gelatin and PVA.This creates a functional wound dressing that combines protection with biological activity.

3.5 FORMULATION EQUATION

GELATIN SOLUTION

$$\text{Gelatin} + H_2O \rightarrow \text{Gelatin}\cdots H_2O \text{ (hydrated gelatin / gel formation)}$$

(Gelatin has groups like –NH, –CO, –OH)

PVA SOLUTION

$$PVA + H_2O \rightarrow PVA \text{ solution}$$

$$–[CH_2–CH(OH)]n– + H_2O \rightarrow –[CH_2–CH(OH)]n– \cdots H_2O$$

PLASTICATION

$$\text{Gelatin} + PVA \rightleftharpoons \text{Polymer network (hydrogen bonding)}$$

THIN TRICOL FLIM

Tridax procumbens → Bioactive mixture → (Drying) → Solid film

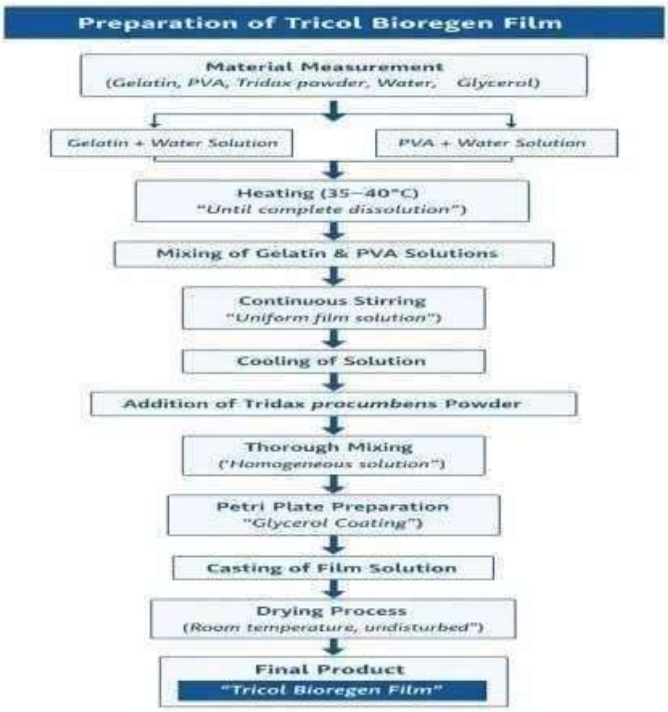


Figure 3.1

Preparation of Tricol thin Flim

3.6 ADVANTAGES

The developed material is simple and easy to prepare, making it highly suitable for basic laboratory and practical applications. It does not require any synthetic additives, which enhances its natural composition and reduces the risk of harmful side effects. Additionally, the overall production is cost effective, making it accessible for widespread use, especially in low resource settings.

The material is biocompatible and safe for human use, ensuring that it does not cause irritation or adverse reactions when applied to the skin. Due to these properties, it is highly suitable for wound healing applications, as it supports the natural healing process while providing protection.

Furthermore, the incorporation of herbal extracts from Tridax procumbens enhances the functional efficiency of the material by providing antimicrobial and anti-inflammatory benefits. These properties help in preventing infections, reducing pain and swelling, and accelerating the overall healing process.

The material also maintains a balanced moist environment at the wound site, which is essential for faster cell regeneration and minimal scar formation. Its biodegradable and eco-friendly nature makes it a sustainable alternative to conventional wound dressings

3.7 LIMITATIONS

Despite its many advantages, the developed material also has certain practical limitations that should be acknowledged. One noticeable drawback is its reduced flexibility when a plasticizer is not added. This can make the dressing slightly stiff, which may affect how comfortably it fits over curved or uneven wound surfaces. As a result, handling and proper placement can sometimes be a bit challenging, especially during application on sensitive areas of the body.

Another important factor is the dependence on environmental conditions during preparation. Since the material is dried naturally, variations in temperature, humidity, and airflow can influence the drying time and final texture. For example, in humid conditions, the material may take longer to dry and may not achieve the desired consistency, while in very dry conditions, it might become too brittle .These variations can lead to inconsistencies in performance from one batch to another.

The manual method of preparation also introduces slight variations in the final product. Since the process is not fully automated, differences in mixing, thickness, and extract concentration may occur. This can affect uniformity and reproducibility, which are important factors in biomedical applications. Additionally, the use of natural extracts such as Tridax procumbens may result in limited shelf life, as these bioactive compounds can gradually lose their effectiveness over time if not stored properly.

There is also a possibility of contamination during preparation and storage, especially if strict hygienic conditions are not maintained. Compared to commercially manufactured dressings, the mechanical strength of this material may be slightly lower, which could limit its use in more severe wounds or those requiring long-term durability. Overall, while the material performs well for basic wound care, further improvements in formulation, preservation, and standardization would help enhance its reliability and expand its practical applications.

3.8 CONCLUSION

Overall, the methodology adopted in this study demonstrates a simple yet effective approach for developing a bioactive wound dressing film. By thoughtfully combining natural and synthetic polymers with a medicinal plant like Tridax procumbens, the work successfully balances mechanical strength with therapeutic benefits. This integration allows the material to not only act as a protective layer but also actively support the healing process through its biological properties.

A key highlight of this study is the use of a lubrication-based technique in place of conventional chemical plasticizers. This approach makes the preparation process more straightforward, safer, and environmentally friendly, while still maintaining the desired flexibility and quality of the final product. It reduces dependency on synthetic additives and aligns well with the growing interest in natural and sustainable healthcare solutions.

Further more, the methodology is practical and adaptable, making it suitable for both laboratory-scale preparation and potential real-world applications. The use of easily available materials and simple processing steps ensures that the product can be developed even in resource-limited settings. Overall, this study emphasizes that effective wound care solutions can be achieved by blending traditional herbal knowledge with modern material science, offering a promising direction for future biomedical innovations.

CHAPTER 4

IMPLEMENTATION AND EXECUTION

4.1 INTRODUCTION

The implementation stage represents the most crucial phase of the project, where the conceptual design is transformed into a functional and practical product. In this project, the primary objective was to develop a natural wound healing band using extracts obtained from Tridax procumbens leaves, known for their therapeutic properties. Unlike theoretical analysis, this phase required extensive hands- on experimentation, systematic procedures, and continuous refinement to achieve an effective outcome. Each step, from the collection of fresh plant leaves to the preparation of extracts and incorporation into the dressing material, was carried out with precision and care to maintain quality and consistency.

The process began with the proper identification and cleaning of plant materials, followed by extraction techniques such as drying, grinding, and solvent extraction to isolate the active bioactive compounds. These extracts were then carefully integrated into a suitable biocompatible matrix to form the wound dressing. Multiple trials were conducted to optimize parameters such as concentration, texture, absorption capacity, and adhesion of the dressing. Throughout the process, close observation was maintained to ensure that the product exhibited desirable properties like antimicrobial activity, flexibility, and moisture retention.

Safety and effectiveness were key considerations during implementation. The prepared dressing was evaluated for its skin compatibility and ability to support wound healing without causing irritation or adverse reactions. Adjustments were made whenever necessary to improve performance and ensure reliability. This iterative approach helped in refining the product to meet the expected standards of a biomedical application.

Overall, the implementation phase demonstrated that a simple, naturally available resource can be successfully transformed into a valuable healthcare solution through scientific methods and careful execution. It not only validated the practical feasibility of the project but also highlighted the potential of herbal-based materials in developing cost-effective and sustainable alternatives for modern wound care.

4.2 BASE MATERIAL

The process began with the careful collection of fresh leaves of Tridax procumbens, a medicinal plant well known for its wound healing, antimicrobial, and anti-inflammatory properties. To ensure the highest quality, the leaves were gathered from a clean and uncontaminated environment, away from roadside pollution, industrial areas, or chemically treated fields. This precaution was important to avoid the presence of dust, toxins, or microorganisms that could negatively affect the purity and effectiveness of the final product.

During collection, only young, fresh, and healthy green leaves were selected, as they contain a higher concentration of active bioactive compounds. Leaves that were damaged, infected, discolored, or dried were carefully avoided because they may have reduced medicinal value and could compromise the consistency of the extract. Proper selection at this stage ensures better extraction efficiency and enhances the therapeutic potential of the final wound dressing.

After harvesting, the leaves were handled with care to prevent bruising or degradation of their natural components. They were placed in clean, sterile containers or bags to maintain hygiene and prevent external contamination.

. The quality, freshness, and purity of the raw plant material directly influence the effectiveness, safety, and reliability of the final biomedical product. A well-executed collection process ensures that the wound dressing developed later will possess optimal healing properties, making this step a critical foundation for the success of the entire project.



Figure:4.1

Tridax Procumbens Leaf

4.3 PROCEDURES AND IMPLEMENTATION

4.3.1 Cleaning, Sterilization and Grinding

Once the leaves of Tridax procumbens were collected, they were subjected to a thorough cleaning process to remove dust, soil, and any adhering impurities. Initially, the leaves were rinsed with clean water to eliminate visible contaminants, followed by washing with distilled water to ensure that no additional minerals or impurities were introduced. This step is essential to maintain the purity of the sample and to avoid interference during the extraction of bioactive compounds.

After cleaning, the leaves were carefully dried under shade at room temperature. Direct sunlight was avoided as it may degrade sensitive phytochemicals present in the leaves. Proper drying helps in removing excess surface moisture, which is important to prevent microbial growth and to improve the efficiency of the grinding process. In some cases, mild sterilization methods such as air drying in a clean environment or brief exposure to controlled conditions may be used to further ensure hygiene.

Once completely dried, the leaves were subjected to grinding using a clean grinder or mortar and pestle to obtain a fine paste or powder. This grinding process plays a crucial role in breaking down the plant cell walls, thereby increasing the surface area for extraction. As a result, the release of active compounds such as flavonoids, alkaloids, and other phytochemicals becomes more efficient. Proper and uniform grinding ensures maximum yield of these beneficial substances, which directly contributes to the effectiveness of the final wound healing formulation.



Figure:4.2

Collection of tridax Procumbens

4.3.2 Filtration and Purification

After extraction, the mixture obtained from Tridax procumbens leaves contained both dissolved bioactive compounds and unwanted solid residues such as fibrous materials and plant debris. To obtain a clear and usable extract, filtration was carried out using clean filter paper. The mixture was carefully poured through the filter, allowing the liquid portion to pass through while retaining the solid particles. This step is essential to improve the clarity and quality of the extract.

Following filtration, the collected solution was allowed to stand undisturbed for some time to enable sedimentation. During this process, any fine particles that might have passed through the filter gradually settled at the bottom of the container. The clear supernatant liquid was then gently separated without disturbing the settled impurities. This additional purification step further enhanced the cleanliness and consistency of the extract.

In some cases, repeated filtration or the use of finer filter media can be applied to achieve higher purity. Maintaining a clean and purified extract is very important because the presence of impurities can affect the texture, stability, and performance of the final wound dressing. A well- purified extract ensures a smooth formulation, better absorption, and improved therapeutic effectiveness, ultimately contributing to the overall quality and reliability of the biomedical product.



Figure:4.3

Extraction of Leaf

4.3.3 Freeze Drying (Lyophilization)

The purified extract obtained from Tridax procumbens was further processed using freeze drying, also known as lyophilization, to convert it into a stable powder form. In this technique, the liquid extract was first frozen at a very low temperature, causing the water present in it to solidify. After freezing, the material was subjected to a low-pressure (vacuum) environment, where the frozen water directly sublimates from ice to vapor without passing through the liquid phase. This gentle drying method helps preserve the structure and activity of sensitive bioactive compound.

One of the major advantages of freeze drying is that it minimizes the degradation of heat-sensitive phytochemicals such as flavonoids, antioxidants, and other medicinal components. Unlike conventional drying methods, which may involve heat and lead to loss of potency, lyophilization ensures that the therapeutic properties of the extract are retained effectively. This makes the final product more reliable and consistent in performance.

The end result of this process is a fine, dry, and lightweight powder that has excellent stability and a longer shelf life. The powdered form is easier to store, transport, and accurately measure during formulation. It also allows for better mixing with polymers or other materials used in preparing the wound dressing film. Overall, freeze drying plays a crucial role in improving the usability, stability, and effectiveness of the herbal extract in the final biomedical application.



Figure:4.4

Lyophilization

4.3.4 Gelatin Addition and Film Formation

The powdered extract obtained from Tridax procumbens was incorporated into a gelatin matrix to form a thin, flexible film. Gelatin was selected as the base material because of its excellent film-forming ability, biocompatibility, and natural flexibility, which are important for wound dressing applications. Initially, gelatin was dissolved in a measured quantity of warm distilled water under continuous stirring to obtain a clear, smooth, and lump-free solution.

Once the gelatin solution was prepared, the extract powder was gradually added in small amounts while stirring continuously. This step ensured uniform dispersion of the herbal extract throughout the solution, preventing the formation of clumps and maintaining consistency. Proper mixing is crucial, as it directly influences the uniform distribution of bioactive compounds in the final film, ensuring consistent therapeutic effects across the dressing.

After achieving a homogeneous mixture, the solution was carefully poured or cast onto a clean, flat surface such as a glass plate or petri dish. It was then spread evenly to control the thickness of the film. The casted solution was left to dry under controlled conditions, typically at room temperature, allowing slow evaporation of water. As the solvent evaporated, a thin, flexible, and uniform film was formed.

This dried film acts as the core component of the wound healing band. It not only provides structural support but also serves as a carrier for the herbal extract, enabling direct contact with the wound site. The presence of gelatin enhances the film's flexibility, adhesion, and ease of application, making it comfortable for use on the skin. Overall, this step is critical in transforming the extract into a practical and usable biomedical prod



Figure:4.5

Gelatin Mixture

4.3.5 Testing and Antiseptic Impregnation

Once the film was successfully prepared, it was important to evaluate its overall quality to ensure that it would perform effectively as a wound dressing. The film containing Tridax procumbens extract was carefully tested for key physical properties such as strength, flexibility, and uniformity. A good wound dressing should not be too brittle or too weak; it must be flexible enough to bend along with body movements while still maintaining its structure during use. The film was also checked for its ability to absorb moisture, as maintaining a slightly moist environment is essential for faster and healthier wound healing.

Apart from physical characteristics, the biological performance of the film was also considered. Its antibacterial activity was observed to ensure that it can help prevent the growth of harmful microorganisms at the wound site. This is a crucial factor, as infections can delay healing and lead to complications. The film's interaction with moisture and its stability under different conditions were also noted to confirm that it remains effective during practical use.

After confirming that the film met the required standards, the next step involved integrating it with a sterile gauze material. The film was carefully placed or coated onto the gauze, making it easier to handle and apply directly on wounds. This combination improves usability by providing additional support, better absorption of wound fluids, and proper positioning of the medicated layer.



Figure:4.6

Testing and Antiseptic Impregnation

4.3.6 Final Product Development

The final outcome of this project is the development of the Tricol BioRegen Band, an innovative and natural wound healing dressing formulated using extracts from Tridax procumbens. This product represents the successful transformation of raw herbal material into a practical biomedical application. It is specifically designed to enhance the body's natural healing process by providing a protective barrier over the wound while simultaneously delivering therapeutic benefits such as antimicrobial and anti-inflammatory effects.

The band is lightweight, flexible, and easy to apply, making it suitable for everyday use in both clinical and home care environments. It helps maintain an optimal moist environment, which is essential for faster tissue regeneration and reduced scarring. In addition, it acts as a shield against external contaminants like dust and microorganisms, thereby minimizing the risk of infection. The incorporation of natural bioactive compounds ensures that the healing process is supported in a safe and effective manner.

One of the key advantages of the Tricol BioRegen Band is its safety and eco-friendly nature. Since it is prepared using natural and biodegradable materials, it significantly reduces the chances of skin irritation, allergic reactions, and long-term side effects commonly associated with synthetic products. Its cost-effective production process also makes it highly accessible, especially in rural and low-resource settings where advanced wound care products may not be readily available.

Overall, this product highlights the powerful combination of traditional herbal knowledge and modern scientific techniques. It demonstrates that simple, naturally available resources can be developed into effective healthcare solutions, offering a promising and sustainable alternative to conventional wound dressing methods.



Figure:4.7

Final product

4.4 EXPERIMENTAL SETUP

The successful execution of this project required the use of basic yet essential laboratory equipment at different stages of the process. A grinder or mortar and pestle was used to convert the leaves of Tridax procumbens into a fine paste, which is an important step for efficient extraction of bioactive compounds. Standard glassware such as beakers, flasks, and measuring cylinders were used for mixing, heating, and storing solutions in a controlled and hygienic manner.

A freeze-drying (lyophilization) unit played a key role in converting the liquid extract into a stable powder form without affecting its medicinal properties. This step improved the shelf life and usability of the extract. Drying trays or flat casting surfaces such as glass plates or petri dishes were used for film formation, allowing the mixture to spread evenly and dry into a thin layer.

To maintain proper hygiene and avoid contamination, sterilization tools and practices were followed throughout the experiment. Equipment was cleaned thoroughly, and in some cases, sterilized before use. Maintaining a clean working environment was essential to ensure the safety, purity, and reliability of the final product.

Overall, the coordinated use of these tools and equipment helped in carrying out each stage of the experiment smoothly and effectively, leading to the successful development of the wound healing band.

4.5 PROJECT TIMELINE: Table 2

Process Step	Time Duration	Start Time	End Time
Preparation of Gelatin Base	2 – 3 minutes	0 min	3 min
Formulation of Bioactive Composite	2 – 5 minutes	3 min	8 min
Casting of Antibacterial Layer	1 – 2 minutes	8 min	10 min
Drying and Film Formation	1 – 2 days	10 min	1–2 days
Integration into Bandage Structure	~10 – 15 minutes	After drying	+15 min
Sterilization Process	5 – 10 minutes	After assembly	+10 min

4.6 OBSERVATIONS

During the experimental process, several meaningful observations were recorded that helped improve the quality of the final product. It was noticed that fresh leaves of Tridax procumbens provided better extraction results compared to older or dried leaves, mainly due to their higher content of active compounds. This directly influenced the effectiveness of the wound healing properties in the final formulation.

The freeze drying (lyophilization) technique played an important role in maintaining the stability and potency of the extract. Unlike conventional drying methods, it preserved the sensitive bioactive components,ensuring that the medicinal value was not lost.

The addition of gelatin was also observed to significantly improve the flexibility and strength of the film, making it more suitable and comfortable for use as a wound dressing.

Another important observation was the role of proper mixing. Uniform and continuous stirring ensured that the extract was evenly distributed throughout the gelatin matrix, resulting in a consistent and effective film. Any inconsistency in mixing led to uneven thickness and variation in performance. Overall, these observations highlight how careful material selection and controlled processing conditions are essential for achieving reliable and reproducible results.

4.7 SUMMARY

The implementation phase successfully showed that a natural plant extract like Tridax procumbens can be converted into a useful biomedical product through a well-planned and careful process. Each step, from collection to final product development, was carried out with attention to detail, ensuring quality and effectiveness. The results obtained indicate that the developed material has good potential for supporting wound healing.

CHAPTER 5

RESULT AND DISCUSSION

5.1 INTRODUCTION

To evaluate the effectiveness, safety, and suitability of the developed Tricol BioRegen Band, a series of essential laboratory tests were conducted. These tests are designed to ensure that the material not only promotes wound healing but also remains biocompatible and safe for human use. The three primary tests performed include the Antimicrobial Test, pH Test, and Hemolysis Test, each addressing a critical aspect of wound care performance.

5.2 TESTING PROCEDURES

The Antimicrobial Test is carried out to determine the ability of the developed band to inhibit the growth of harmful microorganisms. Since wounds are highly susceptible to infections caused by bacteria and fungi, this test ensures that the incorporation of Tridax procumbens provides effective natural antimicrobial properties, reducing the risk of infection and promoting faster healing.

The pH Test is conducted to measure the acidity or alkalinity of the band material. Maintaining an appropriate pH level is crucial because the wound healing process is highly sensitive to pH variations. A slightly acidic to neutral pH is generally ideal, as it supports cell regeneration and prevents microbial growth, ensuring a favorable environment for healing.

The Hemolysis Test is performed to evaluate the biocompatibility of the material with blood. This test checks whether the band causes any damage to red blood cells (RBCs) when in contact. A low or negligible hemolysis rate indicates that the material is safe for application on wounds and does not induce any harmful reactions in the bloodstream.

Overall, these tests collectively validate the functional performance, safety, and reliability of the Tricol BioRegen Band as an effective wound healing solution.

5.2.1 Antibacterial Test

The antimicrobial test is performed to evaluate the ability of the Tricol BioRegen Band to inhibit the growth of harmful bacteria, which is essential because infected wounds can delay healing and lead to complications. This test confirms whether the presence of Tridax procumbens in the band provides effective natural antibacterial activity. Typically, the test is carried out using methods like the agar well diffusion technique, where bacterial cultures are spread on a nutrient agar plate and the band sample or its extract is placed onto the surface; the plate is then incubated for a specific period to allow bacterial growth. After incubation, the effectiveness is determined by observing the zone of inhibition. A larger zone indicates stronger antibacterial activity. The results obtained from this test demonstrate that the developed band possesses significant antibacterial properties, making it suitable for protecting wounds from infection and promoting faster healing.

5.2.2 PH Test

The pH test is performed to determine whether the Tricol BioRegen Band has a suitable pH level for wound healing, as an improper pH can cause irritation and slow down the healing process. This test is important because a slightly acidic to neutral pH helps in preventing bacterial growth and supports tissue regeneration. The test is usually done by dissolving or soaking a small portion of the band in distilled water and then measuring the pH using a pH meter or pH paper. The result obtained indicates whether the material is safe for skin application; if the pH falls within the acceptable range, it confirms that the band provides a favorable and non-irritating environment for effective wound healing.



Figure 5.2

PH Testing

5.2.3 Hemolysis Test

The hemolysis test is conducted to check the blood compatibility of the Tricol BioRegen Band, ensuring that it does not damage red blood cells when it comes into contact with blood. This is important because any material used for wound dressing must be non toxic and safe for direct application. In this test, red blood cells are mixed with the sample extract and incubated for a certain period, after which the mixture is centrifuged to observe any breakdown of RBCs. The result is determined by measuring the release of hemoglobin in the solution; a low or negligible hemolysis percentage indicates that the material is biocompatible and safe, while high hemolysis would indicate toxicity. The obtained results confirm that the band is safe for wound use and does not cause harmful effects on blood cells.

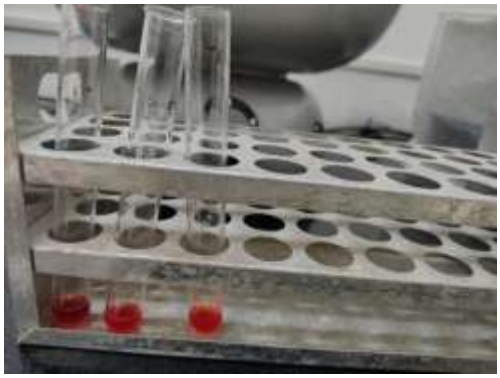
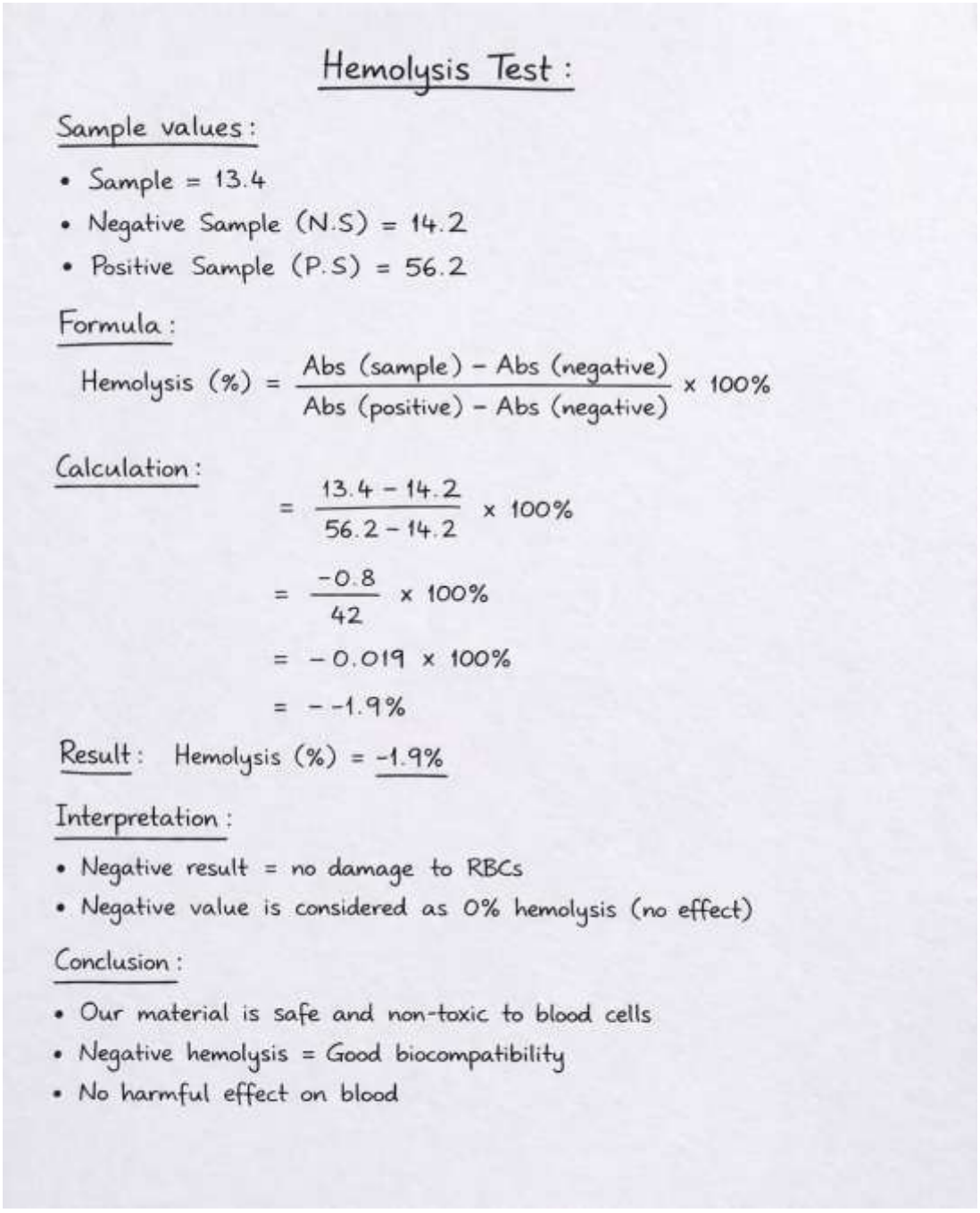


Figure 5.3

Hemolysis Test

Calculation of Hemolysis Test:



Hemolysis Test :

Sample values :

- Sample = 13.4
- Negative Sample (N.S) = 14.2
- Positive Sample (P.S) = 56.2

Formula :

$$\text{Hemolysis (\%)} = \frac{\text{Abs (sample)} - \text{Abs (negative)}}{\text{Abs (positive)} - \text{Abs (negative)}} \times 100\%$$

Calculation :

$$= \frac{13.4 - 14.2}{56.2 - 14.2} \times 100\%$$

$$= \frac{-0.8}{42} \times 100\%$$

$$= -0.019 \times 100\%$$

$$= --1.9\%$$

Result : Hemolysis (%) = -1.9%

Interpretation :

- Negative result = no damage to RBCs
- Negative value is considered as 0% hemolysis (no effect)

Conclusion :

- Our material is safe and non-toxic to blood cells
- Negative hemolysis = Good biocompatibility
- No harmful effect on blood

Figure 5.4

Calculation of Hemolysis Test

5.3 TEST RESULTS

The results obtained from the antimicrobial test, pH test, and hemolysis test indicate that the developed ‘ Tricol BioRegen Band - Nature Boosted Wound Healing using Tridax procumbens with Collagen’ meets the essential requirements for an effective and safe wound dressing material. The positive antimicrobial activity confirms that the band is capable of inhibiting the growth of harmful microorganisms, thereby reducing the risk of infection and supporting faster healing. The pH test results being within the acceptable range demonstrate that the material provides a suitable and non-irritating environment for wound healing. Additionally, the hemolysis test showing negligible or low hemolysis confirms that the band is highly biocompatible and does not cause damage to red blood cells.

Overall, these findings validate that the combination of Tridax procumbens and collagen offers a natural, safe, and efficient solution for wound care. The Tricol BioRegen Band can therefore be considered a promising alternative to conventional wound dressings, with enhanced healing properties and minimal side effects.

5.4 OVERALL DISCUSSION

The project “Tricol BioRegen Band – Nature Boosted Wound Healing using Tridax procumbens and Collagen” was designed to develop a natural and effective wound dressing that not only protects the wound but also actively promotes healing. By combining the medicinal properties of Tridax procumbens with the structural and regenerative support of collagen, the developed band aims to overcome the limitations of conventional bandages that lack antimicrobial and healing enhancing features. The formulation focuses on creating a safe, biocompatible, and skin-friendly material that supports faster tissue repair while minimizing the risk of infection.

The testing results strongly support the effectiveness of the developed band. The antimicrobial test confirmed its ability to inhibit bacterial growth, reducing the chances of wound infection. The pH test showed that the band maintains an optimal and non-irritating environment suitable for healing, while the hemolysis test demonstrated excellent biocompatibility with minimal damage to red blood cells. Together, these results indicate that the Tricol BioRegen Band is a safe, reliable, and efficient wound dressing, making it a promising natural alternative for improved wound care and faster healing outcomes.

CHAPTER 6

CONCLUSION

The project **"TRICOL BIOREGEN BAND": Nature-Boosted Wound Healing using Tridax procumbens with Collagen"** presents an effective approach to wound care by combining natural plant extracts with biocompatible materials. The inclusion of *Tridax procumbens* provides antimicrobial and anti-inflammatory benefits, while collagen supports tissue regeneration and healing.

The developed bandage creates a suitable environment for wound recovery by maintaining moisture and reducing infection risk. Basic evaluations such as absorbency, pH and antimicrobial activity indicate satisfactory performance compared to conventional dressings. Additionally the use of natural and biodegradable materials makes the product cost-effective and environmentally friendly.

Overall, the project demonstrates the potential of integrating herbal medicine with modern biomaterials to develop simple, safe and efficient wound healing solutions.

FUTURE SCOPE

- Development of multilayer bandages for controlled drug release.
- Advanced testing, including in vivo studies and clinical trials, is required to validate the safety and efficiency under real-time medical conditions.
- Improvements in sterilization methods and packaging techniques can help increase product stability and shelf life.

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